Predicate Logic and the Quantifier Rules

"$x^2 \geq 0$ for every real x." "There is no largest prime." "For every $\varepsilon > 0$ there is an N such that $|a_n - L| < \varepsilon$ for every $n \geq N$." The statements of analysis are made of the words "for every" and "there exists", and Chapters 1–3 cannot see them: to a truth table, each of these sentences is a single atom, and the argument "every positive real has a square root; 2 is positive; so 2 has a square root" has the invalid form $p, q \therefore r$. This chapter adds the missing grammar (predicates, terms, and the two quantifiers) and then, exactly as Chapters 1 and 2 did for the connectives, gives the rules **one per section**: how to use a "for every", how to prove one, how to prove a "there exists", how to use one, and the rules for equality. Each quantifier rule comes with a side condition about names, and the side conditions turn out to be exactly the rules that analysts break when a δ secretly depends on an x, or when a witness is treated as if it were arbitrary.

The motivating problem is this. A function f is continuous when for every x and every $\varepsilon > 0$ there is a $\delta > 0$ that works; it is uniformly continuous when for every $\varepsilon > 0$ there is a $\delta > 0$ that works for every x. The two definitions differ only in the order of two quantifiers. *Why does "uniformly continuous implies continuous" have a proof a few lines long, while its converse is false?* The answer is a fact about the order in which names are introduced in a derivation, and Section 4.10 reads it off rule by rule. Along the way the chapter translates mathematical definitions into formulas, negates them quantifier by quantifier, gives formulas a meaning in **models**, and uses models to show that some searches for a derivation must fail. The last section, which is optional, proves that the extended rules are sound.

What you need. Chapters 1–3. From Chapters 1 and 2: the rules for every connective, the definition of a derivation with its boxes and its scope rule (Definition 1.4.2), derived rules (Definition 2.3.1), the classical rules of Section 2.5, and the search method of Section 2.6, which each rule section here extends by one step. From Chapter 3: formulas as objects (Definition 3.1.1), structural induction and recursion (Theorem 3.2.3, Theorem 3.2.6), truth tables, and soundness with its refuting rows (Theorem 3.5.2, Corollary 3.5.3). School arithmetic and the order of $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are used in examples as facts.

Time. About 70 hours on the core track (all of A, one B, and one C per section, the exit checks); about 90 hours for everything.

4.-1.1 Roadmap

- **Predicates and quantifiers.** Predicates, variables, and the domain; how to read "for every" and "there exists", bounded quantifiers, and why the order of two quantifiers can change everything.
- **Terms, formulas, free variables, and substitution.** The exact language, which letters are still holes, the operation "put t in for x" that every rule uses, and the one way it goes wrong: capture.
- **Using "for every": instantiation.** The rule ∀E: a universal statement may be applied to any term, and the term is read off the goal.
- **Proving "for every": let x be arbitrary.** The rule ∀I and the flagged box: prove the statement for a fresh name about which nothing is assumed.
- **Proving "there exists": exhibit a witness.** The rule ∃I, and why the witness is chosen only when an available line shows it.
- **Using "there exists": name a witness.** The rule ∃E, its box, and its fresh-name

condition, and the first derivation that moves an $\exists$ past a $\forall$ (and where the reverse move breaks down).
- **Equality and unique existence.** The rules $=$ I and $=$ E, calculation chains, "there is exactly one", and the completed table of rules, phrases, and proof-assistant commands.
- **Translating mathematics.** Definitions from analysis written as formulas with nested and bounded quantifiers, the bounded forms of the four rules, and the search method in one place.
- **Negating and moving quantifiers.** "Not every" is "some not", negation pushed inward quantifier by quantifier, and the laws that move a quantifier across a connective.
- **Uniformly continuous implies continuous, rule by rule.** The chapter's centerpiece: the full derivation, and the exact lines at which the converse breaks down.
- **Models and counter-models.** A meaning for every formula, in a structure with an assignment; a counter-model as a proof that a sentence does not follow.
- **Soundness for predicate logic** (optional). The soundness theorem of Chapter 3 extended to the quantifier and equality rules, with each side condition used in its own case.

4.-1.2 Named moves

- **Instantiate at the term you need** ($\forall$E): read the term off the goal, not off the order of the premises.
- **Let x be arbitrary** ($\forall$I): a fresh name, about which nothing is assumed.
- **Postpone the witness** ($\exists$I): choose the term only when an available line shows it.
- **Name the witness early** ($\exists$E): "choose x_0 with ...", open the box at once, and never generalize over it.
- **Fix, obtain, choose:** $\forall$I, then $\forall$E and $\exists$E on the hypothesis, then $\exists$I with a term built from what is available. The shape of most proofs in analysis.
- **Negate quantifier by quantifier:** push $\neg$ inward, turning $\forall$ into $\exists$ and back, and keep the bounds.
- **Refute with a model:** a structure in which the premises are true and the conclusion false.
- **Read the dependency off the order:** a variable may depend on the variables quantified to its left, and only on them.